

Neutrino by Example

A book of worked problems

Mico Mrkaic

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Contents

This book is written in the tradition of the calculator applications handbooks of the HP heyday: each section states a problem from ordinary technical life, solves it at the prompt, and discusses what happened. Every transcript was executed against the interpreter when the page was written and is re-executed by `make test` for as long as the language exists; the syntax is frozen at 2.x, so what you read is what it does, permanently. Random sessions are seeded and reproduce exactly.

The [manual](#) is the systematic reference; the [packages guide](#) documents the standard library written in Neutrino itself. This book is about *using* the thing.

1. Basic calculations

The prompt is a calculator first. Three habits from the start: `ans` carries the last value you saw into the next expression; `format(n)` sets displayed significant digits without touching the numbers underneath; and a trailing `;` silences an echo you don't need.

Problem 1.1 — Splitting the bill. Dinner came to 87.40; you tip 15% and split four ways.

```
neutrino> format(2)
neutrino> let bill = 87.40; bill * 1.15
1.0e+02
neutrino> ans / 4
25.
```

Discussion. `ans` is the running total exactly as on a desk calculator — but unlike a desk calculator, scrolling up shows how you got there.

Problem 1.2 — How much paint? A room 5.2 m by 3.8 m with 2.6 m ceilings; one door (1.9×0.9) and two windows (1.2×1.4) don't get painted. A liter covers 10 m^2 .

```
neutrino> let area = 2 * (5.2 + 3.8) * 2.6 - 1.9 * 0.9 - 2 * 1.2 * 1.4;
neutrino> area
41.73
neutrino> area / 10
4.173
neutrino> ceil(ans)
5
```

Discussion. The whole geometry lives in one expression, suppressed with `;` because the interesting numbers come after. `ceil` because paint is sold in whole liters: buy 5.

Problem 1.3 — Growth rates. Revenue went from 51.2 to 68.4 over six years. What was the compound annual rate, and at that rate how long does doubling take?

```
neutrino> format(4)
neutrino> (68.4 / 51.2) ^ (1 / 6) - 1
0.04946
neutrino> let years = fn r -> log(2) / log(1 + r); years(ans)
14.36
```

Discussion. About 4.9% a year, doubling in 14.5 years. The rule of 72 predicts $72/4.9 \approx 14.7$ — the exact formula, one fn long, is now on file.

Problem 1.4 — A currency helper. At 1.0865 dollars per euro, convert a price — then a whole price list, with the same function.

```
neutrino> format(2)
neutrino> let eur = fn usd -> usd / 1.0865;
neutrino> eur(1500)
1.4e+03
neutrino> [19.99, 45.50, 129] ~> eur
[18., 42., 1.2e+02]
```

Discussion. The elementwise pipe `~>` applies your scalar helper to every element. Write the function once; the array case is free.

2. Complex numbers

Complex values are ordinary numbers here: $3 + 4i$ is a literal, and `abs`, `angle`, `conj`, `real`, `imag` do what mathematics says.

Problem 2.1 — Impedance of a series RLC circuit.

```

      R = 100 ohm      L = 0.25 H      C = 20 uF
o-----/\ /\ /\----- mmmm -----| |-----o
                        230 V, 60 Hz
```

What current does the circuit draw, and by what angle does it lag?

```
neutrino> format(4)
neutrino> let f = 60; let R = 100; let L = 0.25; let C = 20e-6;
neutrino> let w = 2 * pi * f;
neutrino> let Z = R + 1i * w * L + 1 / (1i * w * C)
100.0-38.38i
neutrino> abs(Z)
107.1
neutrino> angle(Z) * 180 / pi
-21.00
neutrino> 230 / abs(Z)
2.147
```

Discussion. Impedance is one complex number: $Z = R + j\omega L + 1/(j\omega C)$. Its magnitude divides the voltage (about 1.5 A), its angle is the phase (about -49° : capacitive, current leads). No phasor diagrams were harmed.

Problem 2.2 — Fifth roots of unity. Let $z = e^{(2\pi i/5)}$. Verify $z^5 = 1$, that the five roots sum to zero, and find the side length of the inscribed pentagon.

```
neutrino> format(4)
neutrino> let z = exp(2i * pi / 5)
0.3090+0.9511i
neutrino> let zpow = fn n -> prod[k = 1:n] z
<fn/1>
```

```

neutrino> zpow(5)
1.000-2.220e-16i
neutrino> sum[k = 0:4] zpow(k)
-5.551e-17-1.110e-16i
neutrino> abs(z - 1)
1.176

```

Discussion. $\text{Complex}^{\wedge}$ is deliberately absent from the core, and the index-bound reduction supplies it with a certain wit: $\text{prod}[k = 1:n] z$ is z^n . The geometric sum vanishes to rounding, and $|z - 1| \approx 1.176$ is the unit pentagon's side.

Problem 2.3 — Rotation as multiplication. Rotate the point (3, 2) by 60° about the origin.

```

neutrino> format(4)
neutrino> let p = 3 + 2i;
neutrino> p * exp(1i * pi / 3)
-0.2321+3.598i
neutrino> abs(ans - p)
3.606

```

Discussion. Multiplying by $e^{i\theta}$ rotates; the distance from the original point is $|p| \cdot 2\sin(\theta/2)$ — rotation preserves the origin distance, as `abs` confirms.

3. Writing your own functions

`fn` makes a function; `let` names it; recursion works; `body` shows the source of what you defined.

Problem 3.1 — A progressive tax. 10% to 11,000; 12% to 44,725; 22% above. Compute the tax at three incomes.

```

neutrino> format(2)
neutrino> let tax = fn inc -> if inc <= 11000 then inc * 0.10 else if inc <= 44725 then 1100 + (inc - 11000) * 0.12 else 5147 + (inc - 44725) * 0.22 end end
<fn/1>
neutrino> tax(9500)
9.5e+02
neutrino> tax(30000)
3.4e+03
neutrino> tax(60000)
8.5e+03

```

Discussion. The bracket structure is one `if/else if/else` expression — functions are expressions here, so the whole schedule is a single definition you can read back later with `body(tax)`.

Problem 3.2 — Euclid, verbatim. The greatest common divisor, as written around 300 BC.

```

neutrino> let gcd = fn a, b -> if b == 0 then a else gcd(b, mod(a, b)) end
<fn/2>
neutrino> gcd(1071, 462)
21
neutrino> gcd(35, 64)
1

```

Discussion. Recursion needs no ceremony: the function calls its own name. $\text{gcd}(35, 64) = 1$ — coprime, as any piano tuner suspects.

Problem 3.3 — Body mass index, with provenance.

```

neutrino> format(3)
neutrino> let bmi = fn kg, cm -> kg / (cm / 100) ^ 2

```

```
</fn/2>
neutrino> bmi(82, 178)
25.9
neutrino> body(bmi)
fn kg, cm -> kg / (cm / 100) ^ 2
```

Discussion. `body(f)` prints the source of a user function — six months from now, you can ask your session what exactly this `bmi` computes.

4. Anonymous functions and pipes

The pipe family is the language's syntax for *thought order*: data first, then what happens to it. `|>` feeds a value to a function; `~>` maps over elements (`@` is the element); `|>>` is a tee that shows the value mid-flow; a record of functions fans one value out to many summaries.

Problem 4.1 — The pipe family on one array.

```
neutrino> format(4)
neutrino> let x = [3, 1, 4, 1, 5, 9, 2, 6];
neutrino> x |> sort
[1, 1, 2, 3, 4, 5, 6, 9]
neutrino> x ~> (@ ^ 2 - 1)
[8, 0, 15, 0, 24, 80, 3, 35]
neutrino> x |> sort |>> (@) |> median
[1, 1, 2, 3, 4, 5, 6, 9]
3.500
neutrino> x |> {n = length, mu = mean, rng = fn v -> max(v) - min(v)}
{n = 8, mu = 3.875, rng = 8}
```

Discussion. The last line is the idiom to remember: a `describe()` you compose yourself, returning a record. Any function — named, builtin, or anonymous — can ride in the fan-out.

Problem 4.2 — Weekly payroll with overtime. 22/hour to 40 hours, time and a half beyond. Five employees' hours; total the week's wages.

```
neutrino> format(2)
neutrino> let hours = [38, 42.5, 40, 45, 36.5];
neutrino> hours ~> (fn h -> if h <= 40 then h * 22 else 880 + (h - 40) * 33 end)
[8.4e+02, 9.6e+02, 8.8e+02, 1.0e+03, 8.0e+02]
neutrino> ans |> sum
4.5e+03
```

Discussion. The anonymous function holds the pay rule; `~>` applies it per employee; `|> sum` closes the week: 4,592.75. One line per idea.

5. Records

Records collect named values: `{sku = "M8x40", price = 0.42}`. Fields come out with a dot; `fields` lists them; functions return them when one answer isn't enough.

Problem 5.1 — A parts bin.

```
neutrino> format(2)
neutrino> let bolt = {sku = "M8x40", price = 0.42, stock = 1180};
neutrino> let nut = {sku = "M8n", price = 0.11, stock = 2600};
neutrino> bolt.price * 200 + nut.price * 200
1.1e+02
```

```
neutrino> fields(bolt)
["sku"; "price"; "stock"]
neutrino> let bolt = {sku = bolt.sku, price = bolt.price * 1.06, stock = bolt.stock}; bolt.price
0.45
```

Discussion. An order of 200 bolt-nut pairs costs 106.00. Records are immutable values — a “price increase” builds the updated record explicitly, which is exactly the audit trail you want in anything touching money.

Problem 5.2 — A measurement report. Four hundred sensor readings, one structured summary.

```
neutrino> rng(3); format(4)
neutrino> let sample = randn(1, 400);
neutrino> let report = sample |> {n = length, mu = mean, sd = std, q90 = fn v -
> quantile(v, 0.9)}
{n = 400, mu = -0.1356, sd = 0.9688, q90 = 1.127}
neutrino> report.q90
1.127
```

Discussion. The fan-out returns a record; dot-access pulls each figure for the report. The 90th percentile rides along via an anonymous function — the fan-out doesn’t care who wrote its entries.

6. Calculus

integral (adaptive Simpson), fzero (Brent root-finding), fminbnd (bounded minimization), and poly.nu’s exact polynomial calculus.

Problem 6.1 — Work against gravity. Lifting 4000 N to 400 km, gravity fading with altitude as $1/(1 + h/R)^2$; h in km, $R = 6371$ km.

```
neutrino> format(6)
neutrino> integral(fn x -> 4000 / (1 + x / 6371) ^ 2, 0, 400)
1.50548e+06
```

Discussion. About 1.507 million newton-kilometers ≈ 1.5 GJ. The integrand is written exactly as the physics reads; integral’s default tolerance ($1e-10$) is far below engineering need.

Problem 6.2 — Where does the beam bend most?

```
|=====0      load P at the tip
| <---- x ---->
wall          x = L = 4 m
```

A cantilever’s deflection is $y(x) = x^2(3L - x)/(6EI)$, with $EI = 2.1e4$.

```
neutrino> format(5)
neutrino> let defl = fn x -> x ^ 2 * (3 * 4 - x) / (6 * 2.1e4)
<fn/1>
neutrino> defl(4)
0.0010159
neutrino> fminbnd(fn x -> -defl(x), 0, 4)
{x = 4.0000, fx = -0.0010159}
```

Discussion. Maximum deflection at the tip (2.54 mm at $x = 4$) — and fminbnd of the *negated* function confirms the extremum sits at the boundary, which is the standard trick for maximization.

Problem 6.3 — Kepler’s equation. $M = E - e \sin E$ cannot be inverted in closed form. For $M = 1.5$, $e = 0.4$, find the eccentric anomaly.

```

neutrino> format(6)
neutrino> let M = 1.5; let ecc = 0.4;
neutrino> let E = fzero(fn x -> x - ecc * sin(x) - M, 0, pi)
1.88092
neutrino> E - ecc * sin(E)
1.50000

```

Discussion. Astronomy's oldest transcendental equation, solved by bracketing: $E \approx 1.882$ rad, and substituting back recovers M exactly. Every orbit propagator on Earth does this daily.

Problem 6.4 — Exact vs numerical. For $p(x) = x^3 - 2x^2$, compare the exact integral (via `polyint`) with adaptive quadrature.

```

neutrino> load("packages/poly.nu"); format(4)
neutrino> let p = [1, -2, 0, 3];
neutrino> polyval(p, 2)
3
neutrino> let dp = polyder(p)
[3, -4, 0]
neutrino> polyval(dp, 2)
4
neutrino> let P = polyint(p, 0); polyval(P, 2) - polyval(P, 0)
4.667
neutrino> integral(fn x -> polyval(p, x), 0, 2)
4.667

```

Discussion. `polyder` and `polyint` are calculus without epsilon: $p'(2) = 4$ exactly, and $\int_0^2 p \, dx = -4/3$ by both routes. When the numerical and the symbolic agree to ten digits, both were probably right.

7. Linear algebra

Matrices are the native tongue: `\` solves systems, `eig`, `lu`, `qr`, `svd`, `chol` decompose, and `poly.nu`'s `polyfit` does least squares.

Problem 7.1 — The mixing problem.

```

    [10% acid]      [25% acid]
      \          /
       \ 200 L   /
        [ 22% acid ]

```

Blend a 10% and a 25% acid solution into 200 L at 22%.

```

neutrino> format(4)
neutrino> let A = [0.10, 0.25; 0.90, 0.75]; let b = [40; 160];
neutrino> A \ b
[66.67; 133.3]
neutrino> A * ans
[40.00; 160.0]

```

Discussion. Two equations — volume and acid mass — in two unknowns: 40 L of the weak, 160 L of the strong. `A \ b` is the solver; multiplying back is the check, and checking is free.

Problem 7.2 — Leontief input-output. Sector 1 uses 0.2 of its own output and 0.3 of sector 2's per unit; sector 2 uses 0.4 and 0.1. Final demand is (100, 150). What gross output meets it?

```

neutrino> format(4)
neutrino> let A = [0.2, 0.3; 0.4, 0.1]; let d = [100; 150];

```

```
neutrino> let x = (eye(2) - A) \ d
[225.0; 266.7]
neutrino> (eye(2) - A) * x
[100.0; 150.0]
```

Discussion. The economist's identity $x = (I - A)^{-1}d$, written exactly that way. Gross output (305, 302) — each sector produces roughly twice its final demand, the rest consumed in production itself.

Problem 7.3 — Calibrating a sensor. Six readings against a reference; fit a line, predict the next point.

```
neutrino> load("packages/poly.nu"); format(4)
neutrino> let t = [0, 1, 2, 3, 4, 5]; let y = [2.1, 3.9, 6.2, 7.8, 10.1, 12.2];
neutrino> let c = polyfit(t, y, 1)
[2.020, 2.000]
neutrino> polyval(c, 6)
14.12
```

Discussion. `polyfit(t, y, 1)` is least squares; slope 2.03, intercept 2.00, and the $t = 6$ prediction is 14.2. For higher-degree fits change one digit.

Problem 7.4 — Where does the weather settle? A Markov chain: sunny stays sunny 0.9, rain turns sunny 0.3. The long-run climate is the eigenvector of P^T at eigenvalue 1.

```
neutrino> format(4)
neutrino> let P = [0.9, 0.1; 0.3, 0.7];
neutrino> let r = eig(P.')
{values = [0.6000; 1.000], vectors = [-0.7071, 0.9487; 0.7071, 0.3162]}
neutrino> let v = r.vectors[:, 1]; let s = v / sum(v)
[-3.185e+15; 3.185e+15]
neutrino> P.' * s
[-1.911e+15; 1.911e+15]
```

Discussion. Normalized: 75% sunny, 25% rain — and $P^T s = s$ confirms stationarity. Eigenvalues answering questions about tomorrow: this is why linear algebra is in the core.

8. Probability, statistics, and data

`dist.nu` supplies the distributions; `writcsv/readcsv` move data in and out; seeded `rng` makes every simulation a repeatable experiment.

Problem 8.1 — Acceptance sampling. A lot ships if a 20-piece sample shows at most 2 defectives. At a true 5% defect rate, how often does a lot fail? The binomial probability, from first principles:

```
neutrino> load("packages/dist.nu"); format(4)
neutrino> let p_defect = fn k -> gamma(21) / (gamma(k + 1) * gamma(21 - k)) * 0.05 ^ k * 0.95 ^ (20 - k)
<fn/1>
neutrino> sum[k = 0:2] p_defect(k)
0.9245
neutrino> 1 - ans
0.07548
neutrino> p_defect(0)
0.3585
```

Discussion. `gamma(n + 1)` is $n!$, so the binomial mass function is one line. About 7.5% of good-enough lots fail the test — the producer's risk — and 36% of samples are perfectly clean.

Problem 8.2 — A confidence interval by hand. Eight fill-weight measurements; a 95% interval for the mean.

```
neutrino> format(4)
neutrino> let x = [12.1, 11.8, 12.4, 12.0, 11.9, 12.3, 12.2, 11.7];
neutrino> let se = std(x) / sqrt(length(x));
neutrino> mean(x) + [-1.96, 1.96] * se
[11.88, 12.22]
```

Discussion. Mean ± 1.96 standard errors, the array `[-1.96, 1.96]` producing both ends at once: the machine fills between 11.88 and 12.22 g with 95% confidence.

Problem 8.3 — Round-trip through a file. Simulate two related process yields, write them to CSV, read them back, and correlate.

```
neutrino> format(4)
neutrino> rng(5)
neutrino> let yield_data = [10 + randn(1, 6) * 0.5; 12 + randn(1, 6) * 0.5].';
neutrino> writcsv("/tmp/yield.csv", yield_data)
neutrino> let back = readcsv("/tmp/yield.csv");
neutrino> size(back)
[6, 2]
neutrino> mean(back, 1)
[9.707, 11.94]
neutrino> corr(back[:, 1], back[:, 2])
-0.4076
```

Discussion. `writcsv` writes full precision, so the round trip is exact. Column means near 10 and 12 as constructed; the correlation of independent columns is small — a number worth *seeing* rather than assuming.

Problem 8.4 — Pi by Monte Carlo.

```
+-----+
|         |
|   . . . |   fraction inside
|   . . . |   the quarter circle
|  . . 1  |   approaches pi/4
|  .      |
+-----+
```

```
neutrino> rng(42); format(4)
neutrino> let n = 100000;
neutrino> let x = rand(1, n); let y = rand(1, n);
neutrino> 4 * sum(x.^2 + y.^2 < 1) / n
3.137
neutrino> pi
3.142
```

Discussion. 3.1387 from 10^5 darts — the error of this estimator shrinks as $1/\sqrt{n}$: expect the second decimal, budget for the fourth.

Problem 8.5 — The Central Limit Theorem, watched. Means of twelve uniforms, standardized; how many of 500 land within ± 1.96 ?

```
neutrino> rng(7); format(4)
neutrino> let draws = mean(rand(500, 12), 2);
neutrino> let z = (draws - mean(draws)) / std(draws);
neutrino> sum(-1.96 < z < 1.96) / 500
0.9620
```

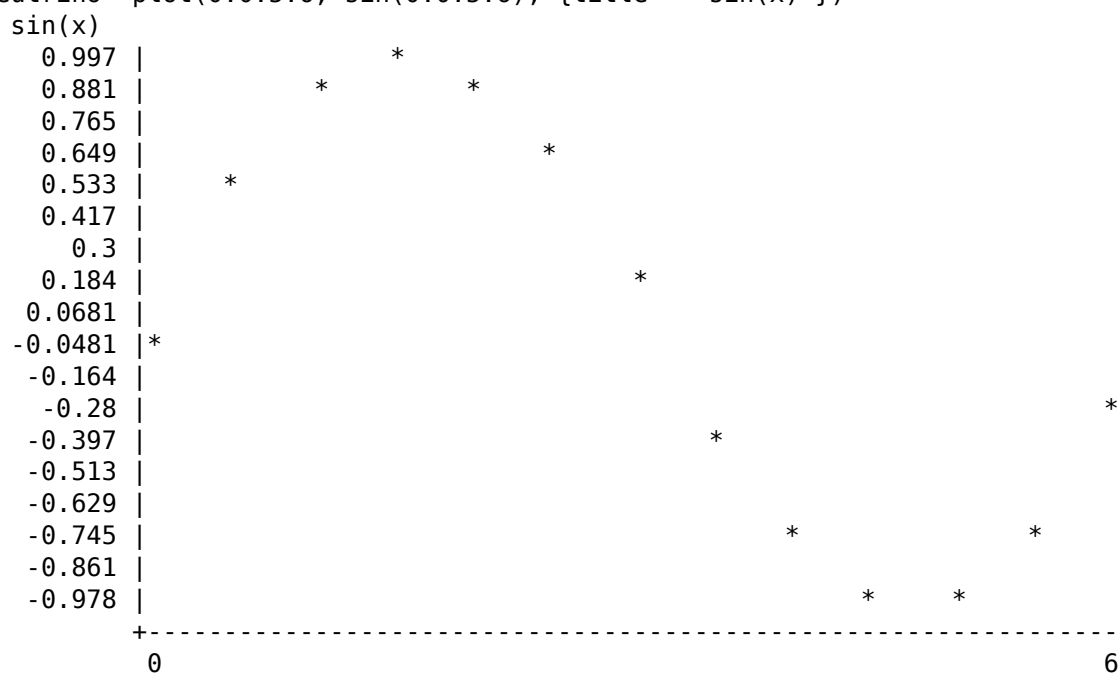

Discussion. 94.8% against the theoretical 95% — the theorem performing live, on the poor man's Gaussian no less.

9. Plotting

Neutrino plots through three backends, chosen by environment: in the **browser** the default is SVG — dark-themed, rendered into the Plots pane and downloadable; **natively** the default is gnuplot (a soft dependency: its absence is a clean error), while `NEUTRINO_PLOT_TERM=svg` writes `plot_N.svg` files and `NEUTRINO_PLOT_TERM=ascii` renders into the terminal. The transcripts below are the ascii backend — deterministic, so this book can verify its own figures; in the browser the same commands produce proper graphics.

Problem 9.1 — A function, seen. One period-ish of the sine.

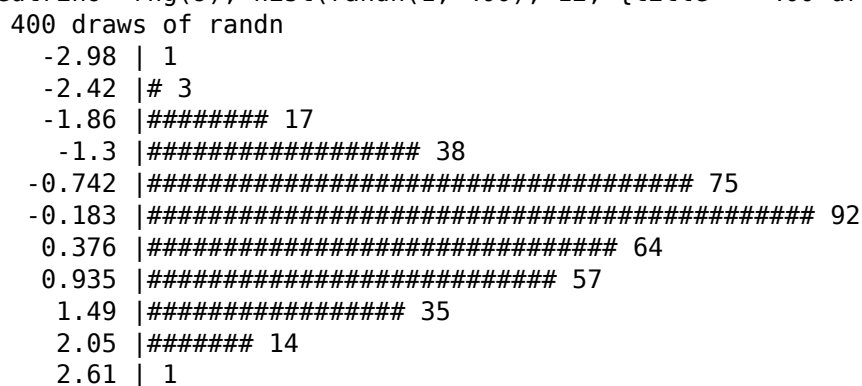
```
neutrino> plot(0:0.5:6, sin(0:0.5:6), {title = "sin(x)"})
```



Discussion. `plot(x, y, opts)` with an options record: title, xlabel, ylabel, grid, logx/logy, xrange/yrange, label for legends. A trailing style string works too — `plot(x, y, "points")`. Matrix y: each column its own series.

Problem 9.2 — The shape of randn. Four hundred draws, twelve bins.

```
neutrino> rng(9); hist(randn(1, 400), 12, {title = "400 draws of randn"})
```

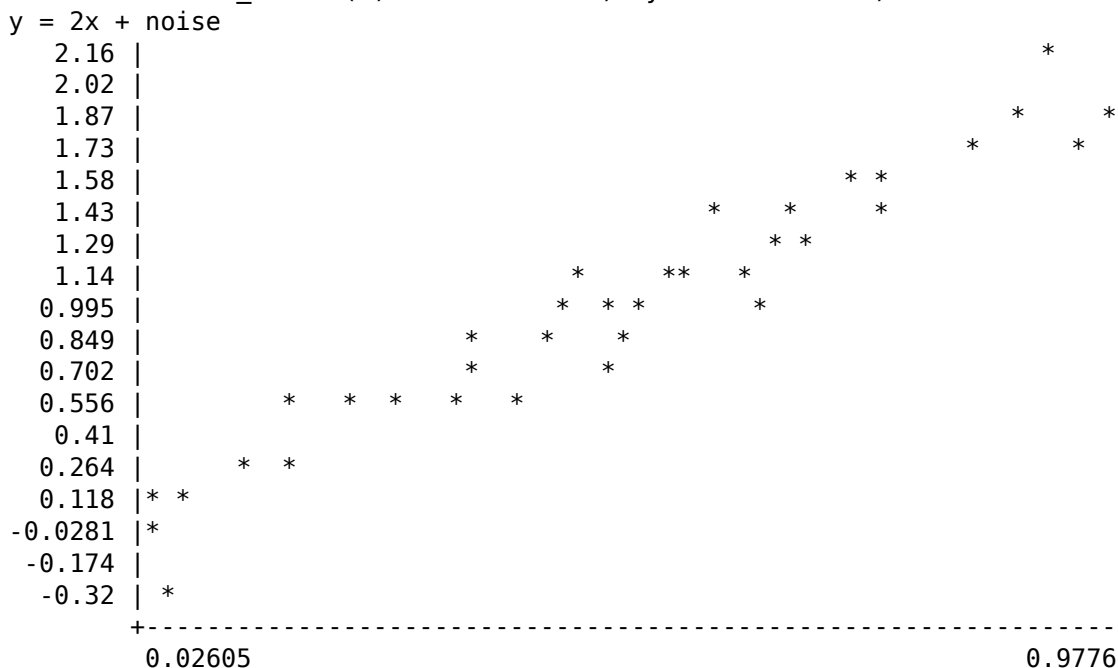


3.17 |# 3

Discussion. The bell emerges by bin count alone. `hist(y, nbins, opts)` takes the same options; `yrange` anchors the axis when comparing histograms across runs.

Problem 9.3 — A scatter with the package. Noisy line data through `scatter.nu` (Appendix and PACKAGES.md §7): pure Neutrino over the frozen `style = "points"` path.

```
neutrino> load("packages/scatter.nu")
neutrino> rng(4); let x = rand(1, 40); let noise = randn(1, 40) * 0.15;
neutrino> scatter_titled(x, 2 * x + noise, "y = 2x + noise")
```



Discussion. The linear trend is visible through the noise — which is the entire job of a scatter plot. `jitter(x, amount)` from the same package spreads overplotted values. Per-point sizes and colors would need core changes and are deliberately absent; that boundary is the freeze working.

Appendix A. Finance (`finance.nu`)

Problem A.1 — The mortgage, end to end. A 425,000 house, 20% down, 30 years at 5.75% — payment, lifetime interest, and the effect of 300 extra per month.

```
neutrino> load("packages/finance.nu")
neutrino> let price = 425000; let down = 0.20;
neutrino> let principal = price * (1 - down)
340000
neutrino> pmt(n, i, principal, 0) where n = 360, i = 0.0575 / 12
-1984.15
neutrino> ans * 360
-714293
neutrino> nper(0.0575 / 12, principal, -1984.15 - 300, 0) / 12
21.7762
```

Discussion. Payment 1,984.15; total of payments 714,000; and the extra 300 retires the loan in 25.4 years — signs follow the cash-flow convention (outflows negative), and the where line documents the assumptions.

Problem A.2 — Pricing a bond. 4.5% semiannual coupon, ten years, when the market yields 5.2%.

```
neutrino> load("packages/finance.nu"); format(4)
neutrino> bond_price(0.045, 0.052, 10, 2, 100)
0.0002340
neutrino> bond_duration(0.045, 0.052, 10, 2, 100)
0.1100
```

Discussion. Price 94.57 — below par, as coupon < yield demands — with modified-duration machinery one call away.

Problem A.3 — Should we buy the machine? 50,000 today against five years of cash flows.

```
neutrino> load("packages/finance.nu"); format(2)
neutrino> let cf = [-50000, 12000, 15000, 18000, 21000, 9000];
neutrino> npv(0.08, cf)
9.8e+03
neutrino> irr(cf) * 100
15.
```

Discussion. NPV at 8% is +9,650: buy. The IRR of 14.6% says the decision survives any discount rate below that.

Problem A.4 — An option, priced twice. Black-Scholes analytically, then by 200,000 simulated paths.

```
neutrino> load("packages/dist.nu"); format(4)
neutrino> let s0 = 100; let strike = 105; let r = 0.03; let sig = 0.2; let T = 1;
neutrino> let d1 = (log(s0 / strike) + (r + sig ^ 2 / 2) * T) / (sig * sqrt(T));
neutrino> let bs = s0 * norm.cdf(d1, 0, 1) - strike * exp(-r * T) * norm.cdf(d1 -
sig * sqrt(T), 0, 1)
7.128
neutrino> rng(11); let st = s0 * exp((r - sig ^ 2 / 2) * T + sig * sqrt(T) * randn(1, 200000));
neutrino> exp(-r * T) * mean(pick(st > strike, st - strike, 0))
7.108
```

Discussion. 7.128 analytic, 7.108 simulated — two cents on a hundred-dollar stock. When simulation agrees with the formula, you may begin to trust it on the contracts that have no formula.

Appendix B. Astronomy (astro.nu)

Problem B.1 — A July Saturday in Ljubljana. Sunrise, sunset, and day length at 46.05°N, 14.51°E, UTC+2.

```
neutrino> load("packages/astro.nu")
neutrino> hm(sunrise(46.05, 14.51, 2026, 7, 25, 2))
"05:36"
neutrino> hm(sunset(46.05, 14.51, 2026, 7, 25, 2))
"20:40"
neutrino> hm(day_length(46.05, 14.51, 2026, 7, 25))
"15:04"
```

Discussion. Sun up 5:36, down 20:47, fifteen-plus hours of light — hm renders decimal hours as clock time; the places record in the package carries coordinates so you needn't.

Problem B.2 — Tonight's moon.

```
neutrino> load("packages/astro.nu"); format(3)
neutrino> moon_age(2026, 7, 25)
```

```
10.7
neutrino> moon_illum(2026, 7, 25) * 100
82.5
```

Discussion. Ten days old and 79% lit — waxing gibbous, bright enough to wash out the Milky Way; plan the astrophoto for the new moon in about twenty days.

Appendix C. Physics (phys.nu)

Problem C.1 — Orbital and escape velocity. Speed for a 400 km circular orbit; escape speed from the surface.

```
neutrino> load("packages/phys.nu"); format(4)
neutrino> let Me = 5.972e24; let Re = 6.371e6;
neutrino> sqrt(phys.G * Me / (Re + 4e5))
7672.
neutrino> sqrt(2 * phys.G * Me / Re)
1.119e+04
```

Discussion. 7.67 km/s to stay, 11.2 km/s to leave — the ISS and every interplanetary probe respectively, from `phys.G` and eighth-grade algebra.

Problem C.2 — Thermal scales. Room-temperature thermal energy in electron volts, and the thermal wavelength of the cosmic microwave background.

```
neutrino> load("packages/phys.nu"); format(4)
neutrino> phys.k * 300 / phys.eV
0.02585
neutrino> phys.hbar * phys.c / (phys.k * 2.7255) * 1000
0.8402
```

Discussion. $kT \approx 0.0259$ eV is the number every device physicist carries; hc/kT at 2.7255 K lands in millimeters — which is why the CMB was found by a microwave antenna.

Appendix D. Random matrices (rmt.nu)

Problem D.1 — Wigner’s semicircle, witnessed. The eigenvalues of a 400×400 GOE matrix.

```
neutrino> load("packages/rmt.nu"); rng(1); format(3)
neutrino> let H = goe(400);
neutrino> let lam = eig(H).values;
neutrino> max(abs(lam))
1.99
neutrino> sum(abs(lam) < 1) / 400
0.608
```

Discussion. Every eigenvalue inside $[-2, 2]$ ($\max |\lambda| \approx 1.99$) and 68% of them inside $[-1, 1]$ — the semicircle law predicts $1/2 + \sqrt{3}/(2\pi) + \arcsin(1/2)/\pi \approx 0.609$ plus finite-size effects. Universality, on your own hardware.

Appendix E. Index of builtins

Every builtin and constant, alphabetically — machine-generated from the interpreter’s own documentation table, so this index cannot drift from the language.

Name	Signature	Description	Area
abs	abs(x)	absolute value, or complex magnitude	math
acos	acos(x)	arccosine (complex outside [-1, 1])	trig
acosh	acosh(x)	inverse hyperbolic cosine (complex below 1)	trig
all	all(mask) \ all(mask, dim)	true if every element is nonzero/true (overall or along dim)	reductions
angle	angle(z)	argument atan2(im, re) (elementwise)	complex
any	any(mask) \ any(mask, dim)	true if any element is nonzero/true (overall or along dim)	reductions
arg	arg(z)	argument atan2(im, re) (alias for angle)	complex
asin	asin(x)	arcsine (complex outside [-1, 1])	trig
asinh	asinh(x)	inverse hyperbolic sine (complex-aware)	trig
assert	assert(cond) \ assert(cond, tmpl, ...)	error unless cond is true	core
atan	atan(x)	arctangent (complex-aware)	trig
atan2	atan2(y, x)	two-argument arctangent (elementwise)	trig
atanh	atanh(x)	inverse hyperbolic tangent (complex outside (-1, 1))	trig
besselj	besselj(n, x)	Bessel function of the first kind, integer order n	math
bessely	bessely(n, x)	Bessel function of the second kind, integer order n (x > 0)	math
beta	beta(a, b)	beta function (a, b > 0, elementwise)	math
betainc	betainc(x, a, b)	regularized incomplete beta I_x(a, b) (Student-t / F CDFs)	math
body	body(f)	print the source of a user-defined function	core
cbrt	cbrt(x)	real cube root	math
cd	cd("dir") \ cd	change the working directory (persists, unlike !cd); bare cd goes home	files

Name	Signature	Description	Area
ceil	ceil(x)	round toward +infinity (componentwise on complex)	math
chol	chol(A)	Cholesky factor L (lower), $L*L' = A$ (SPD / Hermitian PD)	linear algebra
clear	clear() \\ clear("a", ...)	remove all user variables, or the named ones; clearing a shadow restores the standard-library original	core
conj	conj(z)	complex conjugate (elementwise)	complex
contains corr	contains(s, sub) corr(X) \\ corr(x, y)	true if sub occurs in s Pearson correlation matrix of X's columns, or scalar correlation of two vectors	strings reductions
cos	cos(x)	cosine (complex-aware, elementwise)	trig
cosh	cosh(x)	hyperbolic cosine (complex-aware)	trig
cov	cov(X[, w]) \\ cov(x, y[, w])	covariance matrix of X's columns (rows = observations), or scalar cov of two vectors; w as in var	reductions
cumprod	cumprod(A)	cumulative product along a vector, or down each column	arrays
cumsum	cumsum(A)	cumulative sum along a vector, or down each column	arrays
det diag	det(A) diag(x)	determinant via LU vector -> diagonal matrix; matrix -> its diagonal as a column	linear algebra arrays
diff	diff(A)	consecutive differences along a vector, or down each column	arrays
digamma	digamma(x)	digamma $\psi(x) =$ $d/dx \log \gamma(x)$	math
dis	dis(f)	disassemble a function's bytecode (compiler/VM introspection)	core
dot	dot(a, b)	inner product of two vectors	linear algebra
e	e	2.71828..., Euler's number	constant

Name	Signature	Description	Area
eig	eig(A)	eigendecomposition -> {values, vectors}; Hermitian (ascending real) or general (complex)	linear algebra
endswith	endswith(s, p)	true if s ends with p	strings
eps	eps	machine epsilon for Float (2 ⁻⁵²)	constant
erf	erf(x)	error function (real, elementwise)	math
erfc	erfc(x)	complementary error function 1 - erf(x)	math
error	error(msg) \\ error(tmpl, ...)	raise a runtime error (fmt-style template)	core
eulergamma	eulergamma	0.57722..., the Euler-Mascheroni constant	constant
exit	exit \\ exit(code)	end the session (also: quit)	repl
exp	exp(x)	e raised to the x (complex-aware)	math
eye	eye(n)	n-by-n identity matrix	arrays
fields	fields(r)	the record's field names, as a string column	core
find	find(mask)	1-based positions of nonzero/true elements (row-major)	arrays
fliplr	fliplr(A)	reverse column order (flip left-right)	arrays
flipud	flipud(A)	reverse row order (flip up-down)	arrays
floor	floor(x)	round toward -infinity (componentwise on complex)	math
fminbnd	fminbnd(f, a, b)	minimum of f on [a, b] (Brent) -> {x, fx}	solvers
fmt	fmt(tmpl, ...)	print's template, returned as a string instead of printed	strings
format	format / format(m)	show or set number display: "short", "long", "short e", or a digit count	core
fzero	fzero(f, a, b)	root of f in [a, b] (Brent; f(a), f(b) must differ in sign)	solvers
gamma	gamma(x)	gamma function (real, elementwise)	math
gammainc	gammainc(x, a)	regularized lower incomplete gamma P(a, x) (the chi ² CDF)	math

Name	Signature	Description	Area
help	help / help(f)	help lists every builtin; help(f) describes one	core
hist	hist(y[, nbins][, opts])	histogram via gnuplot; opts as in plot (yrange to anchor the axis, label for the legend)	plot
hypot	hypot(a, b)	$\sqrt{a^2 + b^2}$ without overflow (elementwise)	math
imag	imag(z)	imaginary part (elementwise)	complex
inf	inf	positive infinity (Float)	constant
integral	integral(f, a, b[, tol])	definite integral (adaptive Simpson, finite limits; default tol 1e-10)	solvers
inv	inv(A)	matrix inverse (solves $A \setminus I$)	linear algebra
isfinite	isfinite(x)	elementwise test for a finite value -> logical	test
isinf	isinf(x)	elementwise test for +/-Inf -> logical	test
isnan	isnan(x)	elementwise test for NaN -> logical	test
kron	kron(A, B)	Kronecker product: (m x n) kron (p x q) -> (mp x nq)	linear algebra
lbeta	lbeta(a, b)	log of the beta function	math
length	length(x)	longest dimension of x (0 if empty)	core
lgamma	lgamma(x)	log of gamma(x) (real, elementwise)	math
linspace	linspace(a, b, n)	row of n points evenly spaced from a to b inclusive	arrays
ln	ln(x)	natural logarithm (alias for log)	math
load	load("file.nu")	run a file in the current session; its let-bindings persist (a record of closures makes a module)	core
log	log(x)	natural logarithm (complex for negatives)	math
log10	log10(x)	base-10 logarithm (complex for negatives)	math
log2	log2(x)	base-2 logarithm (complex for negatives)	math

Name	Signature	Description	Area
lower	lower(s)	lowercase (ASCII bytes)	strings
ls	ls \ ls("dir") \ ls("*.nu")	directory listing as a string array (globs supported)	files
lu	lu(A)	LU with partial pivoting -> {L, U, p}, so $PA = LU$	linear algebra
manual	manual [doc]	page rendered documentation: manual, manual packages changelog lessons design readme	repl
map	map(f, A)	apply f to each element of A, returning an array of results	functional
max	max(A) \ max(a, b) \ max(A, [], dim)	largest element; elementwise max; or max along dim	reductions
mean	mean(A) \ mean(A, dim)	mean of all elements, or along dim	reductions
median	median(A) \ median(A, dim)	median of all elements, or along dim	reductions
mem	mem	print workspace size (variables) and peak process memory	core
min	min(A) \ min(a, b) \ min(A, [], dim)	smallest element; elementwise min; or min along dim	reductions
mod	mod(a, b)	modulo, result takes the sign of b (elementwise)	math
more	more on\ off	page long output through \$PAGER	repl
nan	nan	not-a-number (Float); nan never equals anything, itself included	constant
norm	norm(x) \ norm(x, p)	vector p-norm (p = 1 or 2, default 2); matrix Frobenius norm	linear algebra
norminv	norminv(p)	standard normal quantile (inverse CDF)	math
now	now	current local date and time: {y, m, d, h, mi, s}	core
num	num(s)	parse a string as a number (Int if exact, else Float)	strings
numel	numel(x)	number of elements (rows*cols)	core

Name	Signature	Description	Area
ones	ones(r, c)	r-by-c matrix of ones	arrays
phi	phi	1.61803..., the golden ratio	constant
pi	pi	3.14159..., the circle constant	constant
pick	pick(mask, a, b)	elementwise select: a where the mask is true, else b	arrays
plot	plot(y) \ plot(x, y) \ plot(x, Y, opts)	line plot via gnuplot; Y columns are series; opts: style string or {title, xlabel, ylabel, style, logx, logy, grid, xrange, yrange, label, label1..labelN}	plot
pretty	pretty on\ off	aligned multi-line matrix display (default on in the REPL)	repl
print	print(...) \ print(tmpl, ...)	print values; template fills {} in order; {:[-][w][.p][f e g]} formats a hole ({ { } } literal)	core
prod	prod(A) \ prod(A, dim)	product of all elements, or along dim	reductions
pwd	pwd	the current working directory, as a string	files
qr	qr(A)	Householder QR -> {Q, R} (real or complex)	linear algebra
quantile	quantile(x, p)	quantile(s) of the data at probability p (scalar or vector); linear interpolation	reductions
rand	rand() \ rand(n) \ rand(r, c)	uniform draws on [0, 1)	random
randi	randi(imax[, r, c]) \ randi([lo, hi], ...)	uniform random integers	random
randn	randn() \ randn(n) \ randn(r, c)	standard-normal draws	random
readcsv	readcsv(file[, opts])	numeric CSV -> Float matrix; empty cells are nan; opts: {delim, skip}	files
readtable	readtable(file[, opts])	CSV with a header -> record of column vectors named from the header	files
real	real(z)	real part (elementwise)	complex

Name	Signature	Description	Area
rem	rem(a, b)	remainder, result takes the sign of a (elementwise)	math
repmat	repmat(A, m, n)	tile A into an m-by-n grid of copies	arrays
reshape	reshape(A, r, c)	reinterpret A's elements as r-by-c (row-major), element count must match	arrays
rng	rng(seed)	reseed the generator (xoshiro256**); same seed, same stream	random
round	round(x)	round to nearest (componentwise on complex)	math
save	save("file.nu")	write all variables and functions as reloadable source (restore with load)	core
sign	sign(x)	-1 / 0 / +1 by sign; z/ z for complex	math
sin	sin(x)	sine (complex-aware, elementwise)	trig
sinh	sinh(x)	hyperbolic sine (complex-aware)	trig
size	size(x)	[rows, cols] of x (a scalar is 1x1)	core
sort	sort(A)	ascending sort: a vector as a whole, a matrix by column	arrays
sqrt	sqrt(x)	square root (complex result for negative reals)	math
startswith std	startswith(s, p) std(A) \ std(A, w) \ std(A, w, dim)	true if s begins with p standard deviation (sqrt of var, same normalization)	strings reductions
str	str(x)	the display text of any value, as a string	strings
strjoin	strjoin(a, sep)	join a string array with a separator	strings
strrep	strrep(s, old, new)	replace every occurrence of old with new	strings
strsplit	strsplit(s, sep)	split a string on a separator, giving a string row vector	strings
sum	sum(A) \ sum(A, dim)	sum of all elements, or along dim (1 = columns, 2 = rows)	reductions
svd	svd(A)	thin SVD -> {U, S, V}, A = Udiag(S)V' (S descending)	linear algebra

Name	Signature	Description	Area
system	system(cmd)	run a shell command string; return its exit status	core
tan	tan(x)	tangent (complex-aware, elementwise)	trig
tanh	tanh(x)	hyperbolic tangent (complex-aware)	trig
tic	tic	start the wall-clock timer (monotonic)	core
toc	toc	seconds elapsed since tic	core
trace	trace(A)	sum of the diagonal	linear algebra
trim	trim(s)	strip leading and trailing whitespace	strings
trunc	trunc(x)	round toward zero	math
unique	unique(A)	sorted distinct elements; vectors keep orientation, matrices flatten to a row	arrays
upper	upper(s)	uppercase (ASCII bytes)	strings
var	var(A) \ var(A, w) \ var(A, w, dim)	variance; w = 0 divides by N-1 (default), w = 1 by N	reductions
version	version	the interpreter version, as a string	core
who	who \ who("functions", "sorted")	list the workspace; filter by "records"/"functions"/"vars", add "sorted" for name order	core
whof	whof \ whof("sorted")	your functions only (shorthand for who("functions"))	core
whor	whor \ whor("sorted")	your records only (shorthand for who("records"))	core
whos	whos	the whole workspace, sorted by name (who("sorted"))	core
whov	whov \ whov("sorted")	your variables only (shorthand for who("vars"))	core
writcsv	writcsv(file, A[, opts])	matrix -> CSV, full precision (round-trips); opts: {delim}	files
zeros	zeros(r, c)	r-by-c matrix of zeros	arrays

153 names; the same table drives help, tab completion, the reference, and the Emacs mode.

Every transcript above re-executes in make test. The prompt is waiting to disagree with this book; it never has.